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De novo functional transcriptomics with RNA-seq and Ribo-seq

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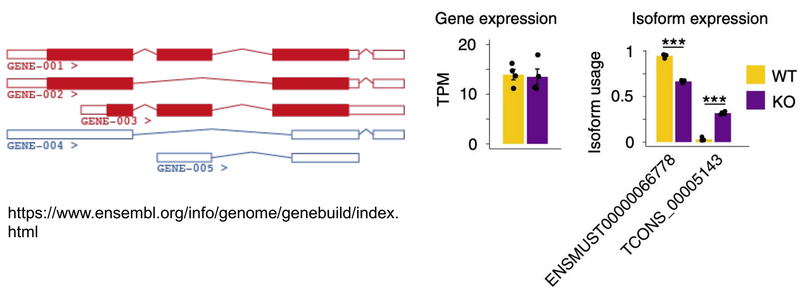
Open problem

Complete and accurate human annotation information about genes and transcripts is still missing.

Differences in expression, function, translational levels and degradation levels that are not detectable at the gene level → Need to study at the isoform level

Moreover, different transcriptomes take part at the different stages of the gene expression regulation cascade (e.g., nuclear retention).

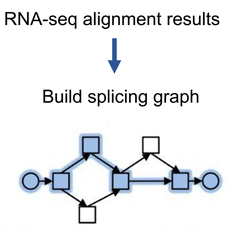


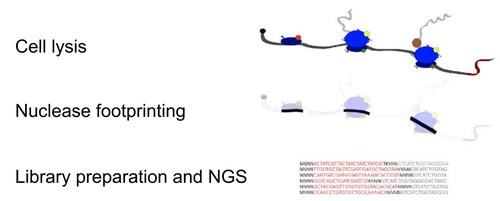
https://www.ensembl.org/info/genome/genebuild/index.html

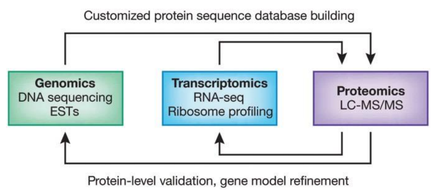
Ringeling et al., *Nature Biotechnology*, 2022

Computational strategies for the characterization of sample-specific transcriptomes, translatomes, and proteomes

Transcriptomics (NGS and third generation seq) ⇄ Ribosome profiling (Ribo-seq) ⇄ Proteomics (mass spectrometry) and proteogenomics







De novo identification of novel and annotated transcripts
Detection and quantification of transcripts by performing mapping-first transcriptome assembly

De novo identification of novel and annotated ORFs
The R package *ORFquant* allows ORF detection and quantification of ORF translation at the level of single isoforms
Moreover, study of translational dynamics and regulatory mechanisms in which translation is involved (NMD, etc.)

De novo identification of novel and annotated proteins
Detection and quantification of peptides and proteins by analyzing MS spectra and by using proteogenomic applications

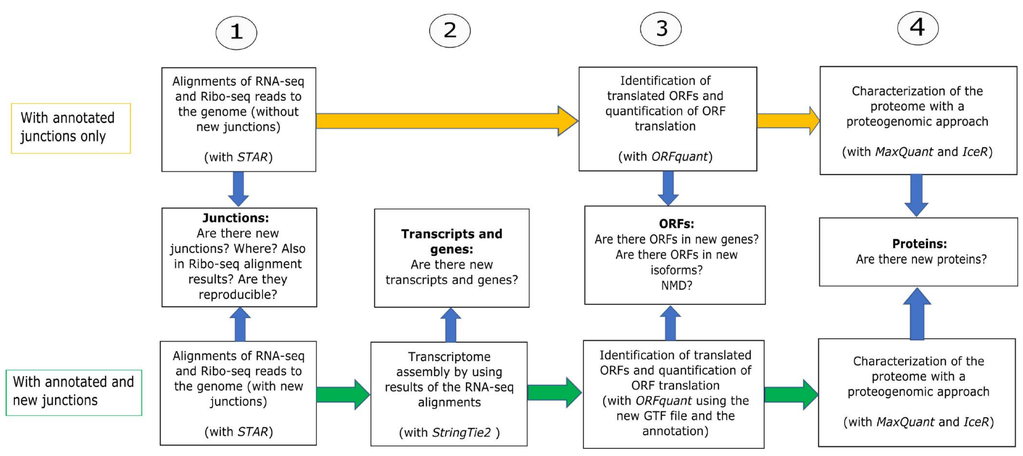
Adapted from M. Pertea et al., *Nature Biotechnology*, 2015
AI Nesvizhskii, *Nature Methods*, 2014

Quantification of the contribution of novel transcripts to the translated transcriptome and evaluation of transcriptome assemblies

We devised and implemented a computational pipeline to study the functions of annotated and novel transcripts.

Moreover, we are working on the development of methods to evaluate and compare transcriptome assemblies by computing measures such as sensitivity and specificity.

	With support	Without support
Novel junction not in the annotation	False Negative (FN)	True Negative (TN)
Novel junction in the annotation	True Positive (TP)	False Positive (FP)



Identification of novel junctions and exonic fragments

Novel junctions were obtained by providing the aligner *STAR* with the RNA-seq data and by running the aligner in two-pass mode.

The final group of junctions can be divided into two main sets:

- 390993 annotated junctions (Junctions which were in the GENCODE v38 annotation)
- 57847 novel junctions (Junctions which were found by running the aligner *STAR* in two-pass mode)

We also devised and implemented a computational strategy to obtain potential novel exonic fragments.

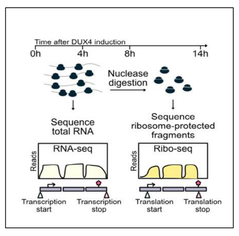
Our workflow allows us to obtain the genomic coordinates of regions that do not overlap with known (GENCODE v38) exons and that contain RNA-seq and/or Ribo-seq signal. The *GenomicRanges* command *reduce* was useful when defining this group of regions, which we collectively call *islands of signal*.

Characterization of condition-specific translated transcriptomes

The *DUX4* dataset: Ribo-seq and RNA-seq data of human skeletal muscle cells at different time points (0, 4, 8, and 14 hours) after the activation of *DUX4*.

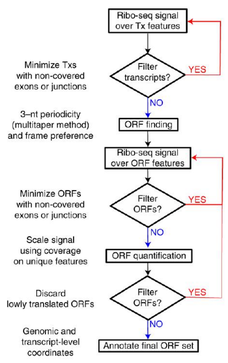
We ran our *de novo* transcript-ORF finding pipeline (*STAR-StringTie2-ORFquant*) and obtained the transcripts and ORFs of each time point.

In this way, we characterized the translated transcriptomes of the four time points.

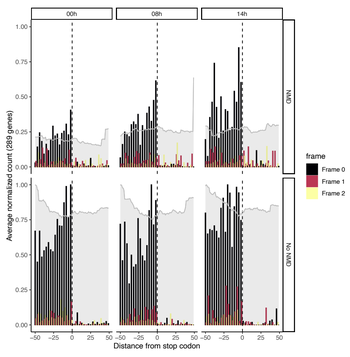


Campbell et al., *Cell Reports*, 2023

On the right, the *ORFquant* workflow (Calviello et al., *Nature Structural & Molecular Biology*, 2020)



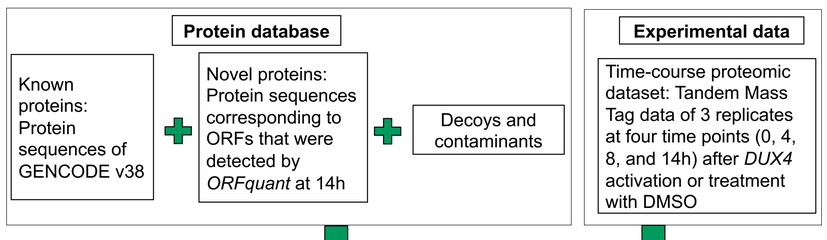
DUX4 misexpression in human skeletal muscle causes facioscapulohumeral muscular dystrophy (FSHD) as well as inhibition of NMD. We therefore investigated whether our Ribo-seq data showed increase in translational levels of transcripts that were NMD targets.



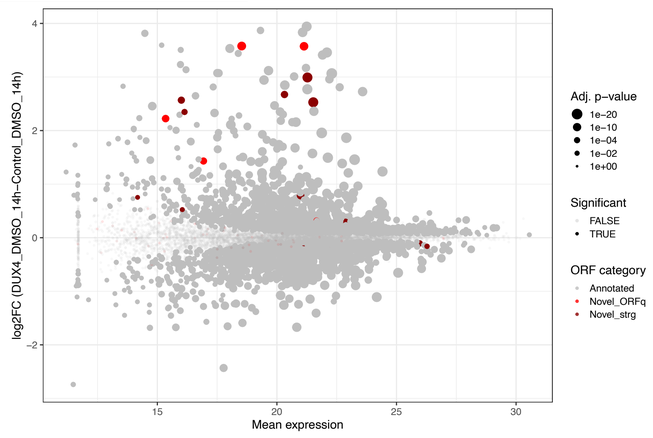
Metaplots of RNA-seq and Ribo-seq profiles around the stop codons of NMD-inducing ORFs (top) and canonical ORFs (bottom) at 0, 8, and 14 h of *DUX4* activation.

Analysis of proteomic data using proteogenomic applications

To characterize the consequences of *DUX4*-mediated misregulation to the proteome of muscle cells, we are analyzing a TMT dataset matching the time points of the transcriptomics assays. We are using a proteogenomic approach to perform our analysis.

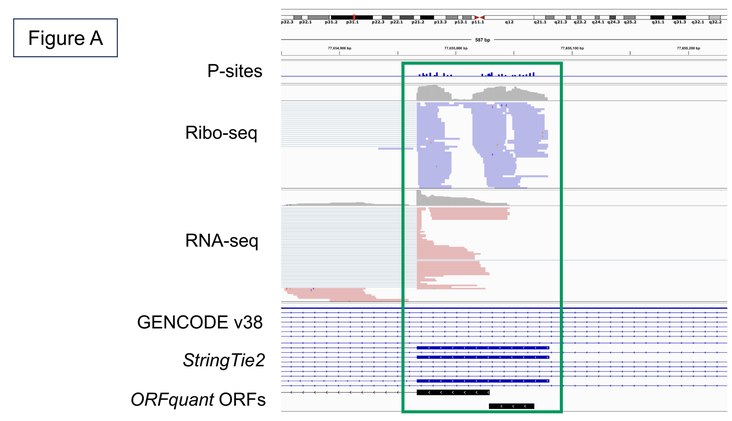


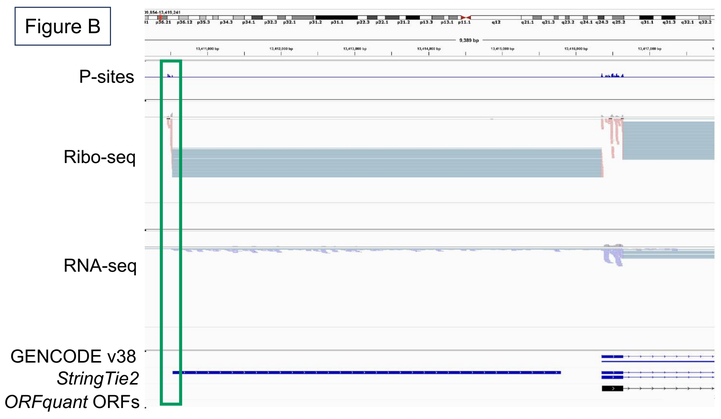
	Down	Not DE	Up
Annotated	1633	7236	1489
Novel_ORFq	1	25	5
Novel_strg	4	23	10



Examples of potential novel exons

We found islands of signal flanking novel junctions which were included in the *StringTie2* annotation objects (Figure A, showing an island/novel exon belonging to a novel isoform) as well as islands flanking novel junctions which were not included in the *StringTie2* annotation objects (Figure B, showing an island/potential novel exon of an isoform).





Future plans

Characterization of condition-specific translated transcriptomes

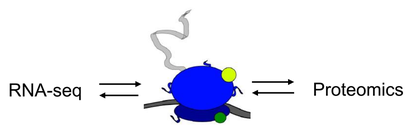
- Devise a better definition of True Positives and True Negatives for transcriptome assembly evaluation
- Study and quantify the contribution of repeat elements to the *DUX4*-induced transcriptome

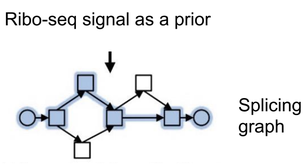
Analysis of proteomic data using proteogenomic applications

- Work on refining the process of protein database construction in terms of content and size
- Perform peptide-level analyses to identify differentially expressed peptides
- Use the same proteogenomic approach for the analysis of other proteomic datasets

Examples of potential novel exons

- We plan to use both novel junctions and exonic fragments as well as annotated junctions and transcript exons to build splicing graphs. In the graphs, exons and exonic fragments will be represented as nodes, whereas edges will be used to represent junctions and to connect adjacent exonic fragments.
- The splicing graphs will be used by a transcriptome assembler that will use Ribo-seq signal as a prior.





Conclusions

Results of this work highlight the potential of computational methods that exploit ribosome profiling data for the detection and characterization of currently unannotated transcripts and ORFs. By jointly using RNA-seq data, Ribo-seq data, and proteomic data it is possible to get deep insights into regulation and functions of the human translated transcriptome in both physiological and pathological conditions.

References

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